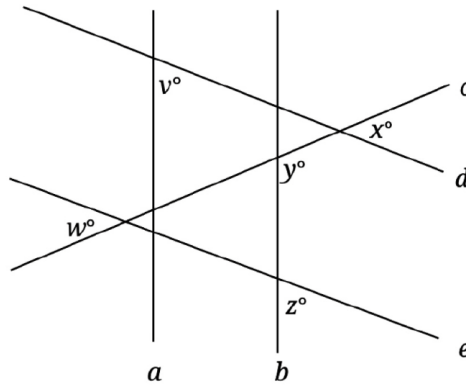


Math Hard Question 10

1

★ Mark for Review

ABC



In the figure, parallel lines a and b are intersected by lines c , d , and e . If $z = 49^\circ$, $y = 136^\circ$, and $v < z$, which statement about x and w must be true?

A. $x < w$

B. $x > w$

C. $x = w$

D. $x + w = 90$

2

★ Mark for Review

ABC

The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines?

I. $f(x) = 16(1.3)^{x+5}$

II. $g(x) = 16(1.09)(1.3)^{x+3}$

A. I only

B. II only

C. I and II

D. Neither I nor II

3

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ABC

The table shows values of x and corresponding values of $g(x)$, where:

$$g(x) = \frac{f(x)}{x+3}$$

and $f(x)$ is a linear function.

What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

x	$g(x)$
-15	3
-6	0
9	5

A. (0, -6)

B. (0, 4)

C. (0, 8)

D. (0, 24)

4

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ABC

The positive number a is 2,106% of the sum of the positive numbers b and c , and b is 81% of c .

What percent of b is a ?

A. 4706%

B. 2600%

C. 38.12%

D. 21.87%

5

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ABC

In the given equation, u is a positive constant. The sum of the solutions to the equation is 1. What is the value of u ?

$$7x^2(3x - 11)(3x - u) = 0$$

A. 1

B. -8

C. $\frac{11}{3}$

D. $\frac{8}{3}$

6

★ Mark for Review

ABC

Which of the following values of x satisfy the equation

$$-3x^2 + 12x + 4 = 0$$

A. 0**B.** -2 and 2 **C.** $2 - \frac{2\sqrt{6}}{3}$ and $2 + \frac{2\sqrt{6}}{3}$ **D.** $2 - \frac{4\sqrt{3}}{3}$ and $2 + \frac{4\sqrt{3}}{3}$ **7**

★ Mark for Review

ABC

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k .

Which of the following could **not** be equivalent to k for all values of x ?

A. $-14x + 1540$ **B.** $14x - 320$ **C.** $-28x + 1720$ **D.** 360

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side.

If a triangle has side lengths of 7 and 10, which inequality represents the possible lengths, x , of the third side of the triangle?

A. $x < 17$

B. $x > 17$

C. $3 < x < 17$

D. $x < 3$ or $x > 17$